

Credit Card Fraud Detection Pipeline

PYTHON 3.13

STREAMLIT CLOUD READY

DOCKER & SQLITE

1. PROJECT OVERVIEW

The **Credit Card Fraud Detection System** is an end-to-end Machine Learning web application designed to evaluate financial transaction risks in real-time. It provides a Streamlit-powered analytical dashboard, automated data pre-processing, classification modeling, and embedded SQLite transaction logging.

The project is engineered for both **containerized local development** (via Docker Desktop and WSL2 resource limits) and **cloud production deployment** on Streamlit Community Cloud.

2. KEY FEATURES

- **Real-Time Inference Dashboard:** Interactive controls to simulate transaction metadata and principal components (V1–V28).
- **Machine Learning Classifier:** Trained model pipeline capable of scoring high-risk fraudulent transactions with probability output.
- **Embedded Persistence:** Automatic transaction recording into an SQLite database (`data/fraud\_predictions.db`).
- **Resource-Optimized Local Docker Execution:** Explicitly calibrated to run smoothly under capped WSL2 memory footprint (4GB RAM limit).
- **Zero-Config Cloud Hosting:** Fully optimized for Streamlit Community Cloud deployment using standard `requirements.txt` manifests.

3. PROJECT DIRECTORY STRUCTURE

```
creditcardFD/
├── .wslconfig           # WSL2 hardware capping (4GB RAM, 2 CPUs)
├── Dockerfile           # Container definition (Python 3.13-slim)
├── docker-compose.yml   # Container service orchestration
├── requirements.txt     # Streamlit Cloud dependency manifest
├── README.md           # Repository README
├── data/
│   └── fraud_predictions.db # Embedded SQLite database
├── models/
│   └── model.pkl         # Serialized ML classifier / preprocessor
└── scripts/
    └── dashboard.py      # Main Streamlit web application entrypoint
```

4. LOCAL DEVELOPMENT & ENVIRONMENT SETUP

Follow these instructions to run the project locally using Python virtual environments or Docker Desktop.

Option A: Running via Python Virtual Environment

1. Clone the repository and navigate into the root directory:

```
git clone https://github.com/ruchitharuchi0575-ux/creditcardFD.git
cd creditcardFD
```

2. Create and activate a Python virtual environment:

```
python -m venv .venv
# On Windows PowerShell:
.\.venv\Scripts\Activate.ps1
```

3. Install dependencies and start the dashboard:

```
pip install -r requirements.txt
streamlit run scripts/dashboard.py
```

Option B: Running via Docker & WSL2 (Stabilized Setup)

Hardware Optimization for Windows 11 / WSL2

To prevent host system freezes during Docker builds, place the following configuration in your Windows user directory (C:\Users\DELL\.wslconfig):

```
[wsl2]
memory=4GB
processors=2
swap=2GB
```

Restart WSL2 and start the Docker container in background-detached mode:

```
wsl --shutdown
docker compose up -d
```

Access the running local app at: `http://localhost:8501`

5. SERVICE MANAGEMENT COMMANDS

Task	Command	Description
Start Container	<code>docker compose up -d</code>	Launches the app on port 8501 in background mode.
Stop Container	<code>docker compose down</code>	Safely stops application and frees port 8501.
Check Memory	<code>docker stats --no-stream</code>	Verifies RAM usage stays within safety bounds.
View Logs	<code>docker logs credit_card_fraud_pipeline</code>	Displays application output logs.

6. CLOUD DEPLOYMENT (STREAMLIT COMMUNITY CLOUD)

The project is configured to deploy directly from GitHub to Streamlit Community Cloud without requiring specialized server infrastructure.

1. Push Code to GitHub

```
git add .
git commit -m "Deploy update"
git push origin main
```

2. Configure Streamlit Form

Set **Branch** to `main` and **Main file path** to `scripts/dashboard.py`.

Dependencies (`requirements.txt`)

Streamlit Cloud reads this exact package list to instantiate the remote Python runtime:

```
streamlit
pandas
numpy
scikit-learn
joblib
```

7. TROUBLESHOOTING GUIDE

Error Symptom	Root Cause	Resolution
<code>This branch does not exist</code>	Stale browser cache on Streamlit Cloud before `git push`.	Press <code>Ctrl + F5</code> on Streamlit Cloud to refresh repository status.
<code>fatal: repository not found</code>	Unconfigured or placeholder remote URL in Git config.	Run <code>git remote set-url origin <actual-url></code> .
<code>Docker daemon disconnected / npipe error</code>	WSL2 memory exhaustion during container build.	Configure <code>.wslconfig</code> with 4GB limit and run <code>wsl --shutdown</code> .

8. LICENSE & AUTHORS

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